

Solution Requirements

Date	03 July 2026
Team ID	SWTID-2026-9325
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system must allow borrowers to securely register and log in using an email and an encrypted password via JWT (JSON Web Tokens) to manage protected user sessions.</i>	High
2	Authorization levels	<i>The platform must restrict access so that authenticated standard borrowers can only view, create, or modify their own specific loan records, financial profiles, settlement histories, and AI-generated letters.</i>	High

3	External interfaces	<i>The backend application must securely connect to the external Google Gemini AI API via official SDKs/endpoints to pass borrower financial payloads and retrieve intelligent strategies.</i>	High
4	Transactions processing	<i>The FastAPI services must process core data exchanges safely by executing full CRUD operations on database entities (Users, Loans, Financial Profiles, Settlement Records, and AI History) utilizing SQLAlchemy ORM.</i>	High
5	Reporting	<i>The platform must generate a responsive financial dashboard showing real-time health metrics, calculated debt-to-income (DTI) ratios, debt stress levels, and historical AI interaction tracking lists.</i>	Medium
6	Business rules, etc.	<i>The financial processing engine must calculate loan settlement priority levels dynamically based on metrics such as outstanding balances, monthly incomes, expenses, monthly surpluses, and overdue durations.</i>	High

7	Compliance to laws or regulations	<i>The solution must ensure financial data privacy protection standards are upheld, handling sensitive borrower debt metrics safely through local system security and isolated storage constraints.</i>	Medium
8	Other	<i>The platform must provide an automated text formatting system that processes raw LLM recommendation strings into clean, downloadable, and copy-ready lender negotiation request letters.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system must load user dashboards swiftly and process backend database operations with minimal delay, excluding external AI API response times.</i>	<i>FastAPI request processing and database query latency under 200ms; frontend page rendering under 2 seconds.</i>
2	Scalability	<i>The application database schema must support efficient data handling as records expand, designed to easily transition from SQLite to production-grade cloud engines if usage increases.</i>	<i>Seamless migration capability to enterprise systems (like PostgreSQL/MySQL) without altering core database entity structures.</i>

3	Security & Data Privacy	<i>The system must protect borrower accounts and stored credentials against unauthorized entry or extraction vulnerabilities.</i>	<i>Enforced password hashing using secure algorithms and complete encryption protection for JWT authorization payloads.</i>
4	Reliability & Availability	<i>The local development system must handle exceptions gracefully without crashing the core server or corrupting the SQLite database file during peak loads.</i>	<i>Successful execution of server operations with zero app downtime during automated local validation and testing procedures.</i>
5	Usability & Accessibility	<i>The React.js frontend interface must be responsive, clean, and intuitive for borrowers who are dealing with high financial stress.</i>	<i>Form validations must prevent layout shifts, and standard web browser layouts must scale cleanly across varied display monitors.</i>
6	Other	<i>The project environment configuration must stay lightweight, isolated, and highly portable to allow other developers to deploy it instantly.</i>	<i>The system environment must be successfully initialized and launched on any standard PC using Python 3.11+, a virtual environment (venv), and a single requirements.txt file.</i>